XX-000

Persistent Dealer-Gamma Regimes Under Temporal Obfuscation: A Re-Run on Complete Inputs with a Controlled LLM Re-Test

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Abstract

Temporal obfuscation tests whether a large language model (LLM) **reasons about numbers or recalls market history**: dates and tickers are removed from 30-day windows of daily dealer gamma exposure (GEX). Our earlier work reported a persistent regime in **every SPY window of 2024–2025**. We re-run the test on the complete data, because the published 2024–2025 inputs **lacked the put leg**. On the complete series the deterministic rule detects **12.0%** of those windows and gpt-6-luna 12.5%. The model follows the rule in 99.7% of 4,691 windows, dates change 1 of 223 verdicts, and an open-weights model run locally gives the same rates.

Introduction

Dealer gamma exposure (GEX) sums the gamma of open options, with the dealer side imputed by a sign convention; a sign that holds for weeks marks a **regime**.

- **Obfuscation** removes dates and tickers: the model has only the values and the criteria, not its memory of the period.
- A re-run needs the **same inputs** and a **model that still answers**: o4-mini is retired and the daily GEX table was rebuilt in 2026, so we recovered the published inputs from the stored batch file.

Research Questions

RQ1 Which **daily series** were the published inputs, and how do the rule's labels change on the **complete series**?

RQ2 Does a current model reproduce the **published answers**, and does its call follow the **numbers**, not the **dates**?

RQ3 Do the answers hold without **0DTE**, in other years and assets, and with **open-weights models** run locally?

Materials and Methods

Data

Daily net GEX of SPY, QQQ and IWM, 2008–2026, from stored option chains, in six series: the published inputs, the call leg only, the pre-rebuild archive, the complete series with and without same-day expiries (0DTE), and an intraday snapshot.

Windows and rule

- 30 trading days, stride 1, labeled **Day T−29 … T+0**; no dates, no tickers; 38,592 windows.
- **Rule:** ≥ 70% of days of one sign, mean |GEX| ≥ $5B, ≤ 5 sign flips.

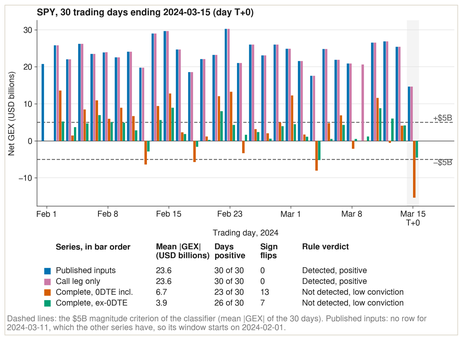
Models

- **o4-mini:** the published answers (January 2026), now retired.
- **gpt-6-luna:** OpenAI Batch API; all its runs cost USD 10.02.
- **gpt-oss-20b, Llama 3.1 8B:** open weights at fixed revisions, vLLM on local A100 GPUs.

Statistics

Cohen's κ for agreement; Wilson and quarter-block bootstrap intervals, since the windows overlap. Every input, prompt and answer is stored.

Approach: Same Rule, Same Prompt, Different Inputs



Window ending 2024-03-15 in four series (paper Fig. 2).

The rule (each 30-day window)

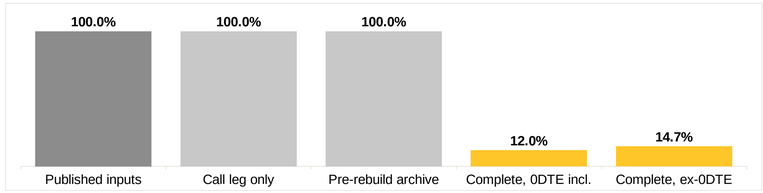
Mean |GEX| < $5B
→ Not detected: below the magnitude floor

< 70% of one sign or > 5 sign flips
→ Not detected: the sign does not persist

All three criteria hold ✓
→ Regime detected, positive or negative

The model gets the 30 values and these criteria only: **no dates, no tickers**.

Same rule, five daily series: share of the 2024–2025 windows that the rule detects



The published 2024–25 inputs carry the **call leg only**, so every day is positive and large.

The **rule** then detects every window.

Both legs: 12.0%. Without 0DTE: 14.7%.

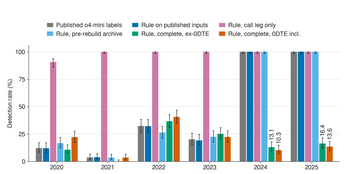
Results

12.0%
2024–25 windows detected
published inputs gave 100%

99.6%
o4-mini = gpt-6-luna
1,412 published windows

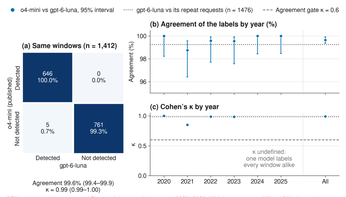
99.7%
gpt-6-luna = rule
4,691 complete windows

1 of 223
verdicts changed by dates
2024 baseline, gpt-6-luna



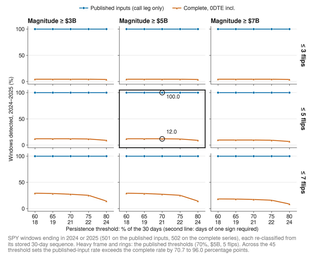
SPY, 30-day windows by year of the window end. Error bars: Wilson 95% CI (overlapping windows treated as independent, so optimistic); a year holds only 9–9 non-overlapping windows. First bar: o4-mini output of January 2026 (parsed answers for 1,412 of 1,476 windows; the 64 answers re-parsed in 2026 are not drawn); other bars: the rule applied to all windows of the series in the legend.

Rule by year: 100% in 2024–25 on the published inputs, 10.3% and 13.6% on the complete series.



SPY windows of the recovered Phase 4A inputs, window ends 2020–2025. Whiskers on years: Wilson 95% intervals, which treat overlapping windows as independent; All: quarter-block bootstrap 95% interval. In (a) the percentage is the share of the row.

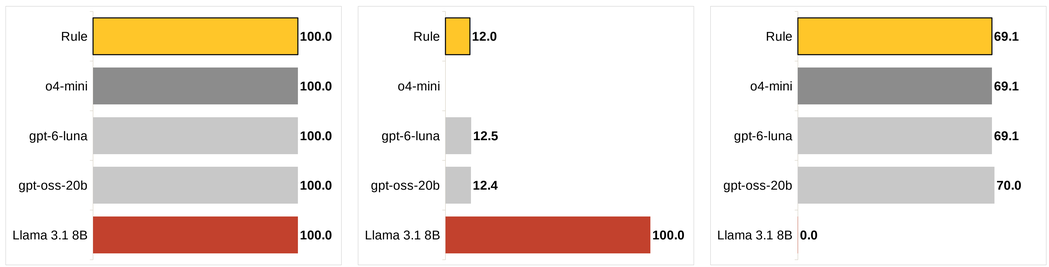
Bridge: gpt-6-luna gives the o4-mini verdict on 1,407 of 1,412 published windows.



SPY windows ending in 2024 or 2025 (501 on the published inputs, 502 on the complete series), each re-classified from its stored 30-day sequence. Heavy frame and rings: the published thresholds (70%, $5B, 5 flips). Across the 45 threshold sets the published-input rate exceeds the complete rate by 70.7 to 96.0 percentage points.

Thresholds: over 45 threshold sets the published inputs stay 70.7–96.0 points higher.

Five labelers compared (o4-mini is retired, so it has no answers on the complete series)



Does It Hold Elsewhere?

Share of windows detected on the complete series:

Check	Windows	Rule	gpt-6-luna
Without 0DTE, 2024–25	502	14.7%	15.3%
2026, after publication	194	11.3%	11.3%
SPY 2008–2013	1,482	0.0%	0.0%
QQQ and IWM	8,569	0.0%	0.0%

Same-day expiries carry **41–46% of |net GEX|** in 2024–26, yet dropping them gives 14.7%, not 100%. Before 2014 and for QQQ and IWM **no window reaches $5B**.

Agreement (Cohen's κ)

κ = 0.993	κ = 0.988	κ = 0.000
o4-mini vs gpt-6-luna	gpt-oss-20b vs rule	Llama 3.1 8B vs rule

Overlap-aware bootstrap, rule 2024–25: **12.0% [5.1, 19.6]**

Conclusions

- **RQ1** ✓ Published 2024–25 inputs: call leg only. Complete series: 12.0%, and no rise after 2023.
- **RQ2** ✓ gpt-6-luna matches o4-mini (99.6%) and the rule (99.7%); dates change 1 of 223 verdicts.
- **RQ3** ✓ Without 0DTE 14.7%, 2026 11.3%, gpt-oss-20b 12.4%. Llama 3.1 8B says "detected" almost always (κ = 0).
- **Next:** thresholds scaled by symbol and price, intraday open interest, more model families.

Take-home: a detection rate is only as sound as its input series; version the data and test a contrast, not a level.

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Code: github.com/iAmGiG/gex-llm-patterns
Repository: scan the QR code

References

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